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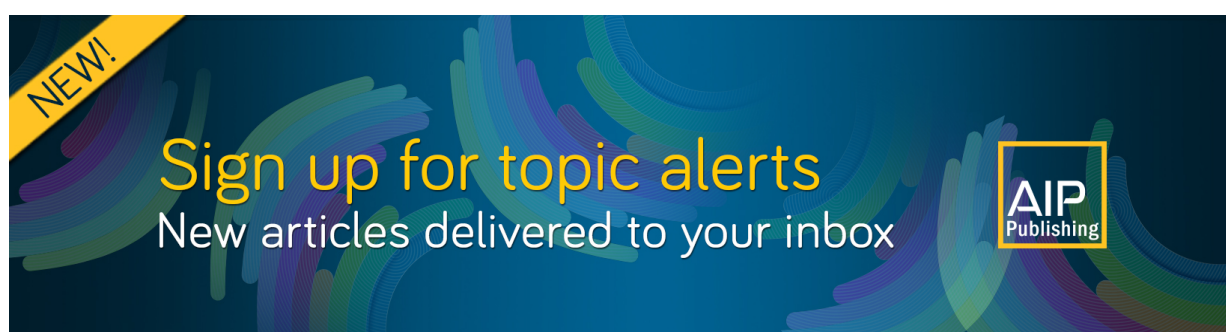
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ABSTRACT

This paper describes the structure of a higher extreme ultraviolet (XUV) emission and plasma density region which surrounds a pinched dense-plasma column. It is interpreted as a toroidal-like plasma formation, which is flowing by a closed current with poloidal and toroidal components upon its surface. This current produces a local magnetic field, which separates the external discharge current from the surface of the dense pinch column. We estimated the values of closed currents as well as magnetic- and plasma-pressures in this column and its surrounding on the basis of the measured distribution of interferometric fringes and intense XUV emission, recorded during the pinch stagnation phase. The considered layer forms the region in which the magnetic energy can be conserved, and during its decay, the acceleration of fast deuterons can take place.

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I. INTRODUCTION

The Z-pinch and plasma-focus (PF) devices^{1–10} are based on the compression of plasma by the magnetic field of the flowing current. This current is non-uniformly distributed upon the surface of the dense pinch column, but it has a complicated structure. The magnetic probes, which were placed in the anode surface of a mega-ampere PF-1000 device,¹¹ recorded a magnetic field in front and back of the dense current sheath. They also recorded the axial components of both orientations.¹² These measurements have led to a conclusion about the existence of closed magnetic fields inside and outside the dense pinch column. This conclusion has been supported by the observed distribution of the plasma density inside the dense pinch column, where two types of plasma structures were observed by means of interferometric diagnostics. The structures of the first type (called plasmoids) had a spherical-like shape with the maximum density in their centers, while structures of the second type had a toroidal-like form

with the maximum density at their internal toroidal axes.^{13–15} The internal structures were explained by the appearance of closed poloidal and toroidal currents and their magnetic fields on the basis of the recorded interferometric fringes and the analysis of their shifts by means of the Abel transformation, as described in earlier papers.^{16,17} It was shown that the plasma density reaches the value of $(10^{24}–10^{25})\text{ m}^{-3}$. An averaged plasma temperature in the considered cases was estimated to reach (50–75) eV. In some quiet phases of the evolution of long-living internal plasma structures, in which we could neglect the kinetic energy density in a comparison with the plasma pressure, we have used the simplifying assumption that the quasi-stationary conditions are achieved locally. In this case, the internal gradient of the plasma pressure ($\text{grad } p$) inside the ordered structures was interpreted by the action of the internal magnetic force which depends on the current density and magnetic field, i.e., $(\mathbf{j} \times \mathbf{B})$. A distribution of the magnetic field in a plasmoid might be similar to that of the

internal currents and magnetic fields in spheromak-like structures, which contain the relaxed toroidal and poloidal currents.¹⁸ Such structures exist at a low β (the ratio of the kinetic and magnetic energy), due to a small internal plasma pressure. In the PF-1000 discharges (at $\beta \approx 1$), the considered dense structures should expand. According to the virial theorem (discussed in a paper¹⁹) at the equilibrium, the mean pressure in the internal plasma region should be balanced by the external pressure. At the pinching plasma, the external pressure is of course produced by the discharge current. The values of the internal current and its magnetic field might be estimated from a plasma pressure gradient. It could be determined on the basis the recorded high-density interferometric fringes, using the Abel transformation of the fringes caused by the observed quasi-symmetrical plasma structures. Using this technique, it was possible to estimate the value of the poloidal current in the plasmoid, and the toroidal current in the toroidal structure, as described in our earlier paper.²⁰

The dense pinch column has usually several sub-regions of a larger diameter (along its length), which are called the lobules. They are distributed symmetrically and/or helically. The distribution of interferometric fringes recorded for the lobules could be explained by an influence of opposite currents which can flow from the lobule tops, also through a rare plasma region.²¹ The internal current constitutes evidently a part of the closed current flowing through the surface of the dense pinch column and the internal boundary of the lobules. It can flow from the lobule tops to the anode. The external current has the direction and value of the main discharge current, and it is pushed by the closed currents to a larger diameter region. The lobule tops were considered as possible sources of fast deuteron emission.^{22,23}

In the present contribution, we broaden the characterization of a layer of the opposite currents in the region outside the dense plasma column and the lobule tops, by a comparison of extreme ultraviolet (XUV) and interferometric frames. We estimate the distributions of the current flow as well as the magnetic- and plasma-pressure. The used experimental device and diagnostics are described in Sec. II. Section III characterizes the structures recorded on the XUV and interferometric frames, and it presents some estimates of the magnetic fields and currents distributions. Section IV presents the summary and conclusions.

II. DESCRIPTION OF THE DEVICE AND DIAGNOSTICS

The experiments were performed using a PF-1000 plasma-focus facility which was equipped with Mather-type coaxial electrodes of 480 mm length. The anode was a thick-wall copper tube of 230 mm diameter, and the cathode was 400 mm in diameter—consisted of twelve stainless-steel tubes (each of 82 mm in diameter). We used a higher filling pressure of pure deuterium (up to 170 Pa) in order to increase the XUV emission and plasma density. The main condenser bank was charged to 16 kV, and the current during the pinch phase amounted to (0.7–0.9) MA. The large dimensions of the electrodes, the higher filling pressure, and the lower energy (stored in the condenser bank) made it possible to observe a relatively slow evolution of the plasma column (and in some phases the quasi-stationary conditions) in a comparison with the optimal parameters needed for the maximal neutron yield.

The voltage and current-derivative waveforms, as well as the total current intensity were measured at the main current collector. Time-resolved soft x-ray (SXR) signals were recorded by means of a silicon

PIN-detector covered with a Be-filter of 10 μm in thickness. Scintillation detector recorded hard x-ray (HXR) and neutron signals at a distance of 7 m from the focus center, in the direction side-on (at 90° to the z-axis). The delay between peaks of the HXRs and fusion-neutrons (of energy 2.45 MeV), as described in an earlier paper,²⁴ was equal to about 300 ns.

Ultraviolet (XUV) photons were recorded by means of a four-frame micro-channel plate (MCP) equipped with four pinholes without any filters. The low-energy limit was about 30 eV, and it was determined mainly by the radiation absorption caused by the working gas in the region between the pinch column and the pinholes. The upper-energy limit was a few keV.

During each PF discharge, we recorded 15 interferometric images of plasma structures at different instants. For that purpose, the use was made of a Mach-Zehnder interferometer equipped with a diagnostics laser (with $\lambda = 527$ nm) generating a 2–3 ns pulse, which was split and delayed (up to 210 ns) by a system of auxiliary mirrors.²⁵ Two half-transparent mirrors of the interferometer were oriented almost parallel in order to obtain wide interferometric fringes. In the recorded interferometric images, the uncertainty was given by the width of these wide fringes of a 1 cm dimension along the (3–5) cm path of the applied diagnostic laser beam. An averaged value of the electron density (n_e), along the path (l) of the laser diagnostic beam of the wavelength $\lambda = 525$ nm, could be calculated from the dispersion formula, as follows:

$$n_e = 4.2 \times 10^{21} \delta / l, \quad (1)$$

where n_e is expressed in m^{-3} , l is expressed in meters, and δ is the number of the shifted interferometric fringes. Hence, we could estimate that the uncertainty of the calculated plasma density in the region of the rarely-spaced fringes (outside the dense pinch column) was below $1 \times 10^{23} \text{ m}^{-3}$.

The Abel inversion computations, which were performed to transform the data from the obtained symmetric interferometric images into the radial density distributions, showed that the electron densities in the toroidal and plasmoidal structures amounted to $(10^{24} - 10^{25}) \text{ m}^{-3}$. Their temporal evolution was discussed in detail in earlier papers.^{16,17} Based on the above data, we could conclude that the dense interferometric fringes confirmed the appearance of high gradients in the electron plasma density, and the dense closed fringes marked the regions of the electron density extremes. It should be mentioned that the angle between the observation axes of the XUV camera and the laser interferometer (in the plane perpendicular to the z-axis) was 45°.

The total neutron yield was measured with two calibrated silver-activation counters. The uncertainty in the timing of different measuring channels during a single discharge was of the order of (3–5) ns. The maximum of the current derivative dip was assigned as the instant $t = 0$.

III. EXPERIMENTAL RESULTS

During the reported experimental campaign, we performed 29 shots at the initial filling pressure of deuterium equal to 170 Pa. That higher pressure enabled a higher plasma density and more intense radiation from the investigated plasma structures to be obtained. That facilitated a better visualization of the layer extending from the tops of the plasma lobules toward the anode. However, the applied filling

pressure (higher than the optimal one) induced a decrease in the total neutron yield the level of 10^9 neutrons per shot. The pictures of the intensively radiating regions were compared with the images of plasma structures visualized by means of the laser interferometry. There were also recorded signals produced by HXRs and fusion neutrons. Some examples, collected when the interferometric and XUV frames were recorded in different phases of the discharge evolution, are presented below.

The waveforms from a shot #12606, which were recorded at the instants close to the appearance of the current-derivative dip, are presented in Fig. 1. It should be noted that the considered discharge produced the total neutron yield equal to about 4×10^9 .

The XUV and interferometric frames from the same shot (#12606) are presented in Fig. 2.

The XUV emission, as presented in Figs. 2(a)–2(d), shows two distinct boundaries: the first—the boundary of the dense plasma column, and the second boundary—extending from the tops of the plasma lobules toward the anode. The partial transparency of this second boundary makes it possible to observe its three-dimensional shape.

The XUV envelope in the region outside the dense column corresponds to an extension of the lobule top in the form of one (or two) narrow fringe(s) visible in the interferometric pictures. In our earlier paper²⁰ it was interpreted as a layer of two counterstreaming currents: the external one—corresponding to the orientation of the main discharge current, and the internal one—flowing toward the anode and constituting a part of the closed current loop upon the surface of the toroidal plasma structure (see the white circles marked in Fig. 2).

Another example of the appearance of the radiating envelope, observed both on the XUV frames and interferometric images, was obtained from the shot #12587 which produced the total neutron yield equal to 2×10^9 . In this case, the XUV and interferometric images were recorded during the period of the plasma stagnation, after the formation of the first plasmoid, but before the evolution of instabilities

and before the dominant fusion-neutrons production, as can be seen in Figs. 3 and 4.

The XUV frames presented in Fig. 4 also demonstrate two distinct boundary layers, similar to those shown above in Fig. 2. The external layer wraps itself around external boundaries of the plasma lobules and extends almost to the anode (gray arrows), while the internal layer embraces the surface of the dense pinch column and internal parts of the lobules. The external boundary emits the XUV radiation also from the layer of a low plasma density imaged in the interferometry frames, and it visualizes the presumed countercurrent layer in this region. The bremsstrahlung depends both on the plasma density and its temperature. It suggests that the local higher temperature can be due to the presence of the current flows.

Considering the interferometric frames, the regions of the current flows can be identified with the axially symmetric regions of the closely-spaced fringes, where the gradient of the plasma density determines (at the similar temperature) the gradient of plasma pressure for the application of the relation $\text{grad } p \approx j \times B$.

The current flowing along the surface of the dense pinch column (seen without internal interferometric fringes), as it follows from the magnetic probe measurements,^{11,12} has a poloidal (black full line) and an azimuthal (black dashed line) component. The orientation of the closed current, as shown in the presented images by the arrows marked upon ellipses, is not ambiguous in this time. The magnetic probes used in PF-1000 experiments^{11,12} showed the orientation of the axial current from the anode, as well as the appearance of an axial magnetic field (equal to about 10% of the azimuthal magnetic field) directed toward the anode and corresponding to some azimuthal current. This orientation was identical in all three shots, when the axial magnetic field was measured, but in front of the imploding plasma and current sheath there was observed a weak B_z field oriented in the opposite direction (i.e., from the anode). The distributions and orientations of the toroidal magnetic fields and poloidal currents have been described in a recently published paper.²⁶ In that case, the magnetic probes recorded the toroidal magnetic field in a compact plasma object ejected from the dense pinch column and observed at a distance of 40 cm in front of the PF-1000 anode. The measurements showed the appearance of the closed poloidal current near the surface and along the z -axis (having the orientation away from the anode). It should be mentioned that the orientation of the toroidal current component was also considered by Bostick in an old paper.²⁷ He interpreted the azimuthal filaments (responsible for the toroidal currents) as the diamagnetic structures, oriented in opposite directions to the magnetic streamlines of the discharge azimuthal magnetic field. This model suggests that the orientation of this B_z component inside the dense pinch column is directed toward the anode. It corresponds to some azimuthal asymmetry of a Z-pinch discharge, which is caused by the direction of the main discharge current flow.

In our present considerations, i.e., in Figs. 4, 6, and 8, we marked the current orientation in the agreement with the distinct helical curves visible in Figs. 4(b)–4(d). Here, and in some other XUV frames, the poloidal current component could flow at the surface of the dense pinch column in the direction from the anode, and the toroidal current component could have the counterclockwise direction. It should, however, be noted that the orientations of the current components have no influence on the values of the calculated currents and magnetic fields in this paper.

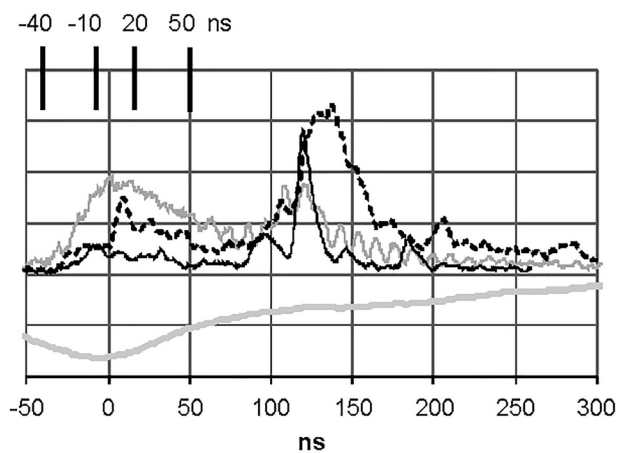


FIG. 1. Waveforms from the shot #12606: the discharge current derivative (thick gray) of the signal from the PIN detector (thin gray), HXRs (thin black), and neutron-produced signals (dashed line) recorded by the side-on detector and shifted in time under assumption that the neutrons had an energy of 2.45 MeV. The numbers above the diagram boundary show the instants when the interferometric images were recorded during the investigated shot.

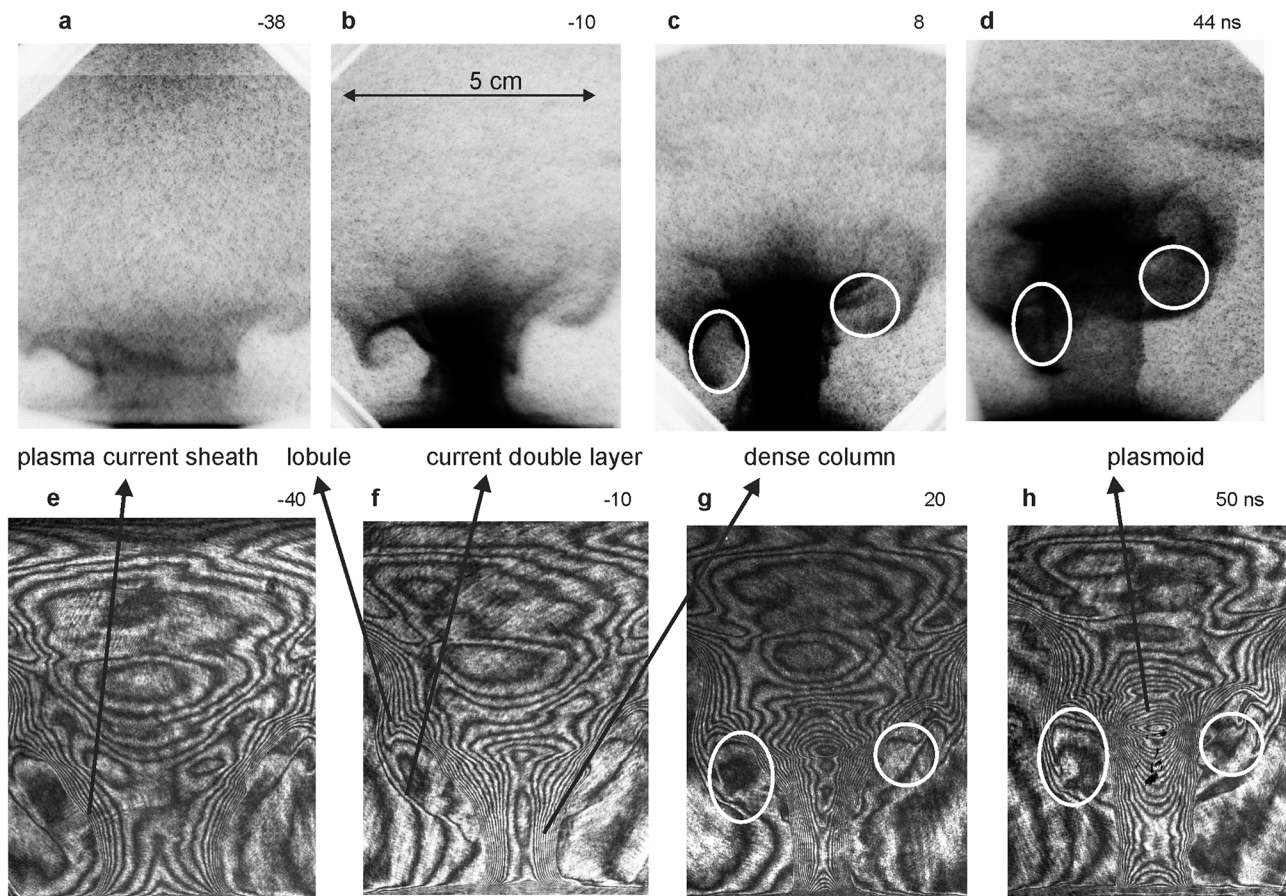


FIG. 2. Images from the shot #12606: XUV frames (top) and interferometric pictures (bottom), which were recorded during the evolution of the first plasmod and the first neutron emission peak. White circles indicate regions of some external toroid-like structures. The arrows show the plasma current sheath, plasma lobule, current double layer, dense plasma column, and an internal plasmod. The bottom edges of the presented images corresponded to the anode face.

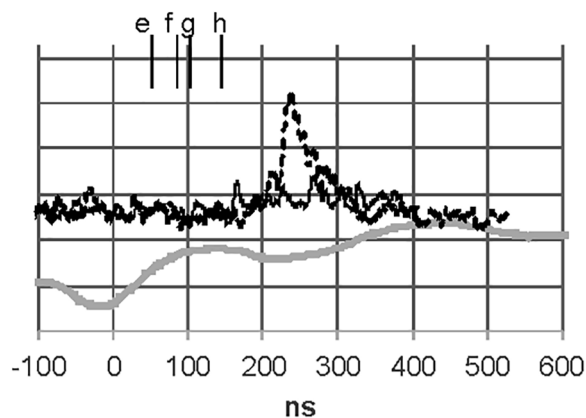


FIG. 3. Waveforms from the shot #12587: the current derivative (thick gray), the HXR signals (thin black), and the neutron-produced signals (dashed line) as recorded by the side-on detector (and shifted in time under assumption that the neutrons had an energy of 2.45 MeV). The letters above the diagram boundary show the instants when we recorded the interferometric images.

It can be concluded that in the shot #12587, the region below the tops of the plasma lobules could conduct (as in the shot #12606) two opposite currents: the external current and the internal current loop. One should also note the regions of a close contact of the both layers visible in the XUV frames 4(a) and 4(c). In Figs. 4(c) and 4(g), this contact is marked by the white circles. It might separate two external closed currents.

One can also observe the expansion of the external boundary, both in the radial and axial directions [see Figs. 4(b) and 4(c)]. This motion might be forced by a magnetic pressure of the increasing closed current flowing partially through the anode. In Fig. 4(h), we recorded a transverse layering of the stagnated dense plasma column, which corresponded to the start of the $m = 0$ instability development.

Another example of the formation of an external current layer was obtained from the shot #12601 which produced the total neutron yield equal to 6×10^8 . In that case, the XUV and interferometric images were taken during the stagnation and start of the compression of the pinch constriction, as shown in Figs. 5 and 6.

In order to support the above conjectures about the presence of the closed currents, we calculated a value of the current flowing along

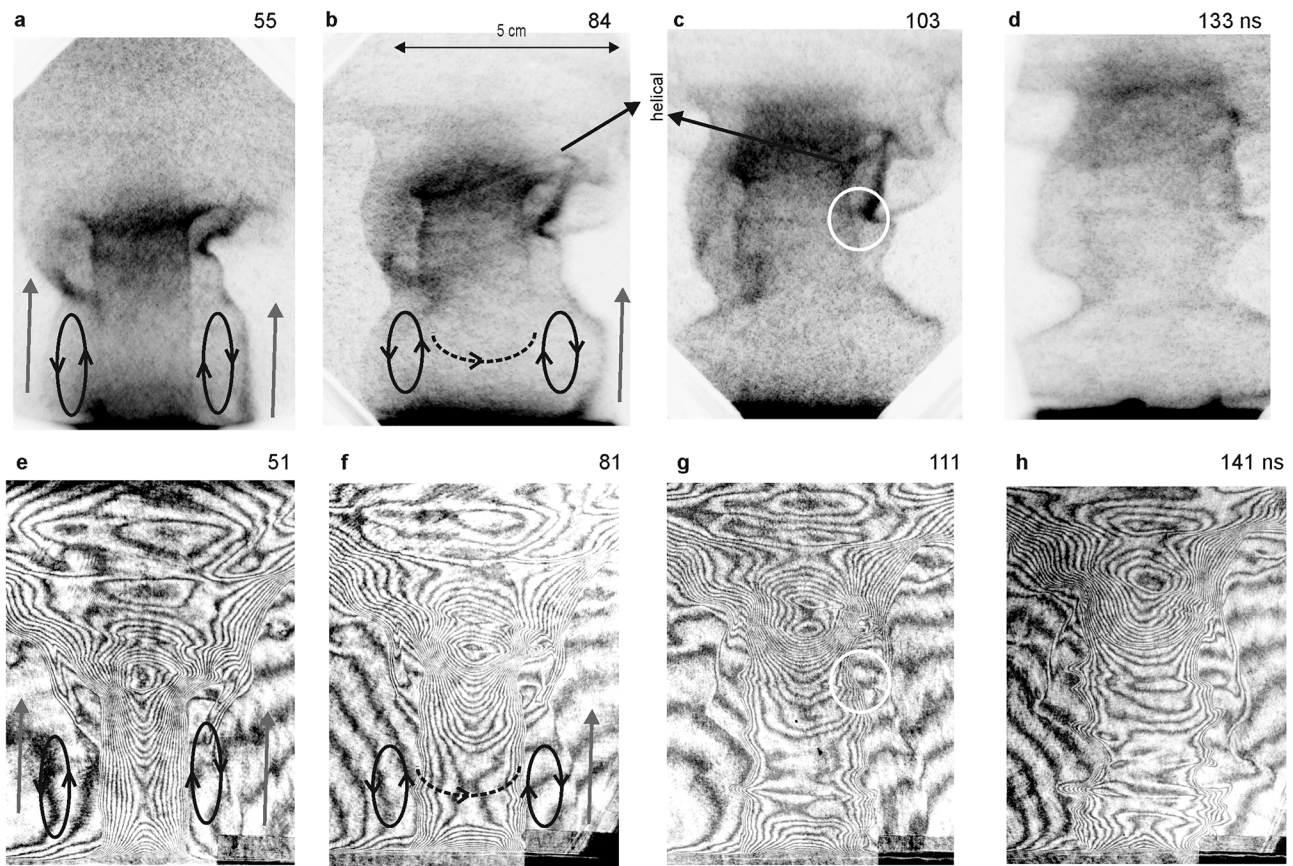


FIG. 4. Images from the shot #12587: XUV frames (a)–(d) and interferometric pictures (e)–(h) recorded during the stagnation phase, which show two radiating layers: the one—upon the surface of the dense pinch column, and the external boundary of the closed current region. The white circle indicates region of the close contact of both layers. The dashed black lines indicate the regions of the considered toroidal currents self-generated around the pinch column, and the full black lines indicate their closed poloidal components. The external current flowing between the anode and cathode is indicated by gray arrows. The helical lines in the frames 4(b) and 4(c) mark the probable orientation of the current flow.

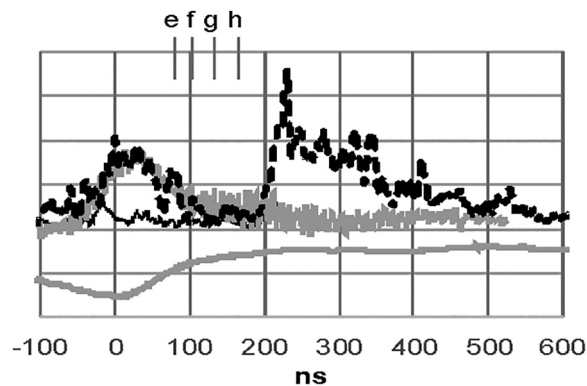


FIG. 5. Waveforms from the shot #12601: the current derivative (thick gray), the signals from the PIN detector (thin gray), the HXRs (black), and signals produced by neutrons (dashed line), measured side-on and shifted in time, as in the previous cases.

the double-layer or the surface of the dense plasma column as shown in Fig. 6(g), in the cross section marked by the black dashed line. It was chosen due to a small asymmetry of the pinch column and the quasi-stationary phase due to a slow evolution of plasma. The calculations were performed at three simplifying assumptions:

1. The intensity of the external pinching current $I_d \approx 700$ kA was estimated on the basis of measurements carried out by means of a Rogowski coil. Its pinching pressure was balanced by the internal pressure of the closed plasma structure according to the virial theorem.¹⁹
2. It was assumed that the plasma temperature T was about 50 eV, as in the previous experiments described in papers,^{16,17,20} and the electron density n_e was about $(10^{24}–10^{25})\text{ m}^{-3}$. The quasi-stationary conditions could be assumed for some quiet pinch phases, as one can see in Figs. 6(e) and 6(h). Outlines of the plasma configuration in the cross section marked in Fig. 6(g) stayed quasi-stable for some time. The plasma transport (which one can observe inside the dense pinch column only) induced the plasma density change to about $4 \times 10^{23}\text{ m}^{-3}$ [see Fig. 7(b)], causing the observed fringe shift by about 5 mm (during 90 ns),

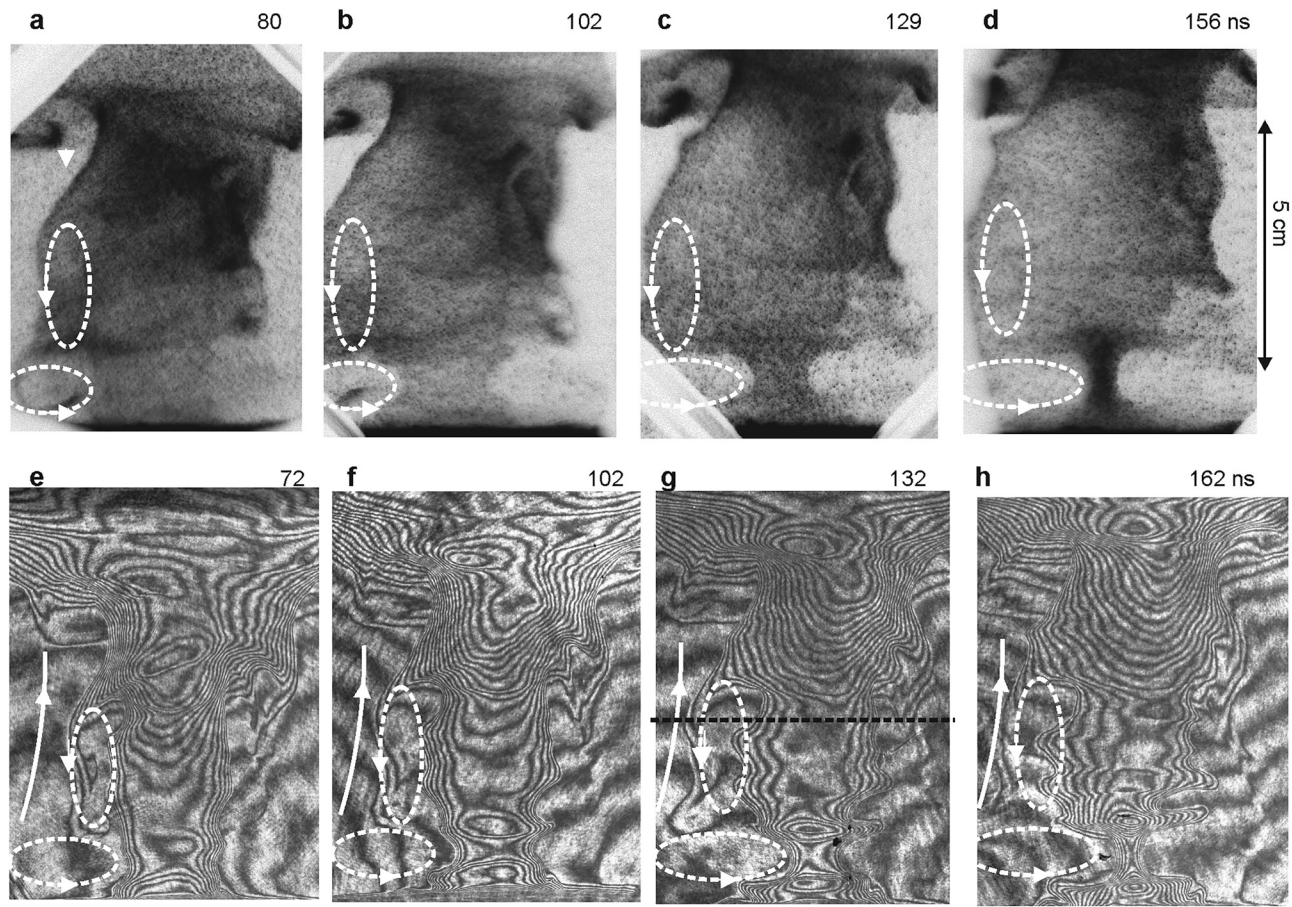


FIG. 6. Images from the shot #12601: the XUV frames [(a)–(d)] and interferometric pictures [(e)–(h)] which show a layer of the intense boundary radiation, a region of the shifted interferometric fringes at the top of the lobule at the external boundary of the closed current loops (poloidal component is marked by white dashed ellipses), and a region of radially displaced external current (white full lines). The black dashed line in the interferometric picture (g) marks the cross section in which the radial distribution of the electron densities, currents, and magnetic pressures was calculated.

- toward the upper plasmoid. In that case, the kinetic energy density amounted to $1/2\rho v^2 \approx 2.0 \times 10^6$ and was about one order lower than the plasma pressure $2n_e kT_e \approx 2.4 \times 10^7$. Therefore, we have conjectured the quasi-stationary conditions locally.
3. The typical velocity of plasma transformation (the Alfvén velocity) had the value of about 10^5 m/s. At the pinch stagnation, the radial dimension of the column did not change considerably [see Figs. 2(c), 2(d), 2(g), 2(h), 4(c), 4(d), 4(g), and 4(h)], then the radial velocity could be neglected. The rotation velocity of the dense column of the order of Alfvén velocity should reach 1 cm during 100 ns. In the surface layers visible in Figs. 2 and 4, one can easily see some asymmetry of the left and right boundaries (both in the XUV and interferometric images). However, the character of this asymmetry did not change during the considered period, so we could not deduce any rotation of the dense plasma column. The velocity of a $(\mathbf{E} \times \mathbf{B})$ drift (at $E < 10$ kV/10 cm and $B \approx 10$ T) should be about 10^4 m/s, i.e., about one order lower than the Alfvén velocity. Consequently, we could assume an equilibrium of the compressing magnetic

pressure, and the repulsive magnetic and plasma ($p_n \approx 2n_e kT$) pressures at each radius.

The radius of the dense plasma column was $R_1 \approx 8$ mm, that of a double layer was $R_2 \approx 15$ mm, and the radial position of the center of the toroid-like structure (important for estimation of the toroidal current and poloidal magnetic field) was assumed to be $R_0 \approx 11$ mm. The subscript 1 was used for the dense column, 2—for the double layer, and 0—for the toroidal structure. The poloidal I_p and toroidal I_t current components in the considered plasma structure were associated with the toroidal B_t and poloidal B_p components of the magnetic field, respectively.

In order to calculate a radial distribution of the electron density, taking into account some asymmetry of the plasma column in the cross section shown in Fig. 6(g), we counted the fringes on the left- and right-part of the recorded interferometric image, and we calculated their averaged number vs the radius, as shown in Fig. 7(a).

On the basis the data presented in Fig. 7(a), using the known Abel procedure, we could calculate the radial distribution of the

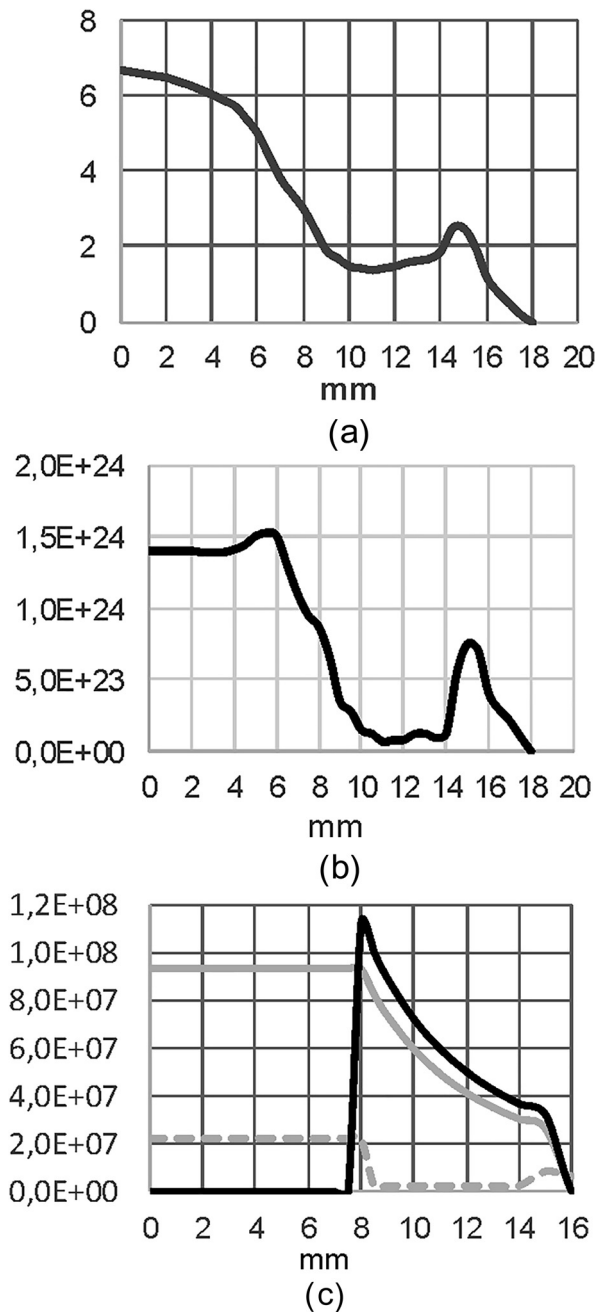


FIG. 7. Results of the computations performed for the cross section marked in Fig. 6(g): (a) the radial distribution of the interferometric fringes along the line of the cross section; (b) the radial distribution of the electron density calculated from the Abel transformation of the data shown in the diagram (a); and (c) the distribution of pressures: the magnetic pressures associated with the pinching poloidal current I_p (black line), the repulsive pressure of toroidal current I_t (full gray), and the plasma pressure p_n (dashed gray).

electron density, as shown in Fig. 7(b). The mean electron density n_e in the tube of the radii equal to (8–15) mm—was $0.1 \times 10^{24} \text{ m}^{-3}$, n_e in the double layer—was about $0.5 \times 10^{24} \text{ m}^{-3}$, and that inside the pinch column (of the radius ranging up to 8 mm)—amounted to about

$1.4 \times 10^{24} \text{ m}^{-3}$. Hence, the plasma pressures reached the values $p_{n0} \approx 0.2 \times 10^7 \text{ Pa}$, $p_{n2} \approx 0.8 \times 10^7 \text{ Pa}$, and $p_{n1} \approx 2.2 \times 10^7 \text{ Pa}$, respectively. The radial distribution of these pressures is presented in Fig. 7(c).

A scheme of the considered currents, magnetic fields, and magnetic pressures is shown in Fig. 8.

During the consider shot, for $I_d \approx 700 \text{ kA}$, the pinching magnetic field at R_2 was $B_d \approx \mu I_d / (2\pi R_2) \approx 9.33 \text{ T}$, and the pinching magnetic pressure was $p_d = B_d^2 / 2\mu \approx 3.47 \times 10^7 \text{ Pa}$. That pinching pressure was equal to the sum of the repulsive plasma pressure $p_{n0} \approx 0.2 \times 10^7 \text{ Pa}$ and $p_{p2} \approx 3.27 \times 10^7 \text{ Pa}$. We have also assumed the pressure balance on the surface of the dense pinch column of the radius $R_1 \approx 8 \text{ mm}$. At this radius, the pinching pressure p_{p2} increased proportionally to $(R_2/R_1)^2$ and amounted to $p_{p1} \approx 11.5 \times 10^7 \text{ Pa}$. This pressure was equal to an increase in the plasma pressure $\Delta p_n \approx 2.0 \times 10^7 \text{ Pa}$ and a repulsive pressure of the toroidal current component $p_{t1} \approx 9.5 \times 10^7 \text{ Pa}$. Comparing both pressures, we could calculate the toroidal and poloidal components of the local magnetic fields: $B_{t1} \approx 17.0 \text{ T}$ and $B_{p1} \approx 15.4 \text{ T}$. Those values of the magnetic fields determined the values of currents in the tube: $I_p = B_{t1} \cdot 2\pi r / \mu \approx 680 \text{ kA}$, and $I_t = B_{p1} \cdot 2r / \mu \approx 270 \text{ kA}$. It should be noted that the value of I_t was calculated assuming a radial decrease in the magnetic fields (as $1/r$). Consequently, the decrease in the plasma density inside the dense pinch column during the stagnation was accompanied by an increase in the toroidal current component from the initial value of about 70 kA to the final one of 270 kA.¹²

The uncertainty of the electron density value was estimated from the uncertainty of the Abel transformation, caused by the observed asymmetry of the plasma pinch column. The value of 20% corresponded to the uncertainty of the electron density determination in the axial region. The dominant uncertainty in our calculations of the magnetic pressure was caused by estimations of the temperature. The upper limit of temperature, i.e., the value of (100–200) eV, was determined from the recorded of H- and He-like lines of C- and N-ions during the SXR peak, observed at the current derivative dip.^{15,16} It might be produced by the stopping of fast deuterons having kinetic energy equal to about 500 eV. During the pinch stagnation, i.e., 50–100 ns later, the SXR and XUV lines were not recorded. That fact

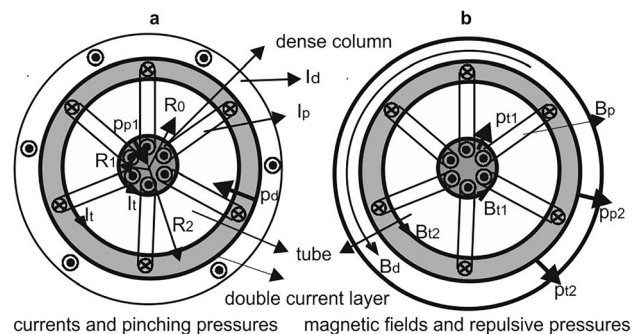


FIG. 8. Scheme (top view opposite to the z-axis): (a) the poloidal and toroidal currents marked as I_p and I_t , and the pinching magnetic pressures marked as p_p and p_t , respectively and (b) the corresponding magnetic fields—marked as B_t and B_p , and the expanding magnetic pressures p_p and p_t , which appear inside and outside the dense pinch column. The spokes represent the poloidal configurations, and the circles—the toroidal ones.

determined that the upper temperature value could be about 75 eV. The lower temperature value was determined by the transmission of the XUV radiation recorded in the XUV frames. It was given by the absorption of the radiation by the filling gas inside the discharge chamber, and it was estimated to be about 30 eV. Therefore, in our calculations, we used a value of 50 eV, with an uncertainty of about 50%. Then the resulting uncertainties of the magnetic field and currents could reach factor two.

Another example of the XUV and interferometric frames, as recorded for later phases of the PF discharge, was obtained from shot #12597, which produced the total neutron yield equal to 3×10^9 . In that shot, the disruption of a distinct pinch constriction and the formation of secondary plasmoids could be observed, as shown in Fig. 9.

The presented images showed a quasi-stable cylindrical boundary of the upper part of the pinch column. In the bottom part of this column, one could observe the evolution of constrictions and the formation of secondary plasmoids. The neutron production corresponded to disruptions of the constrictions and the secondary plasmoids, at a decay of the plasma column and release of the magnetic energy. It should be noted that also in this phase, the external boundary of

toroidal plasma structure could be observed and conserved [see the white ellipses in Fig. 9(e)].

The frames presented in Fig. 9 can be explained also by the formation of a thin toroidal-like plasma structure of higher XUV radiation and denser plasma, which could be a result of the appearance of the poloidal and toroidal current components.

Hence, this paper presents a deeper insight into the magnetic structure of the plasma focus discharge during the pinch evolution. For the phases of relatively slow changes of the fringe patterns, based on x-ray estimates of the local plasma temperature, one may conjecture that the kinetic energy can be an order of magnitude smaller than the thermal pressure. It suggests that the pressure gradient can be almost equal to the magnetic force density. The XUV pictures taken with the unfiltered pinhole camera, as well as the interferometric fringe patterns (recorded with nearly parallel semi-transparent mirrors) showed some details of a low density plasma structure formed outside the dense pinch column. There were recorded two distinct boundaries: the first one which shows the edge of the dense plasma column, and the second one which wraps itself around this region. The considered model takes into account the sign of the estimated pressure gradient as

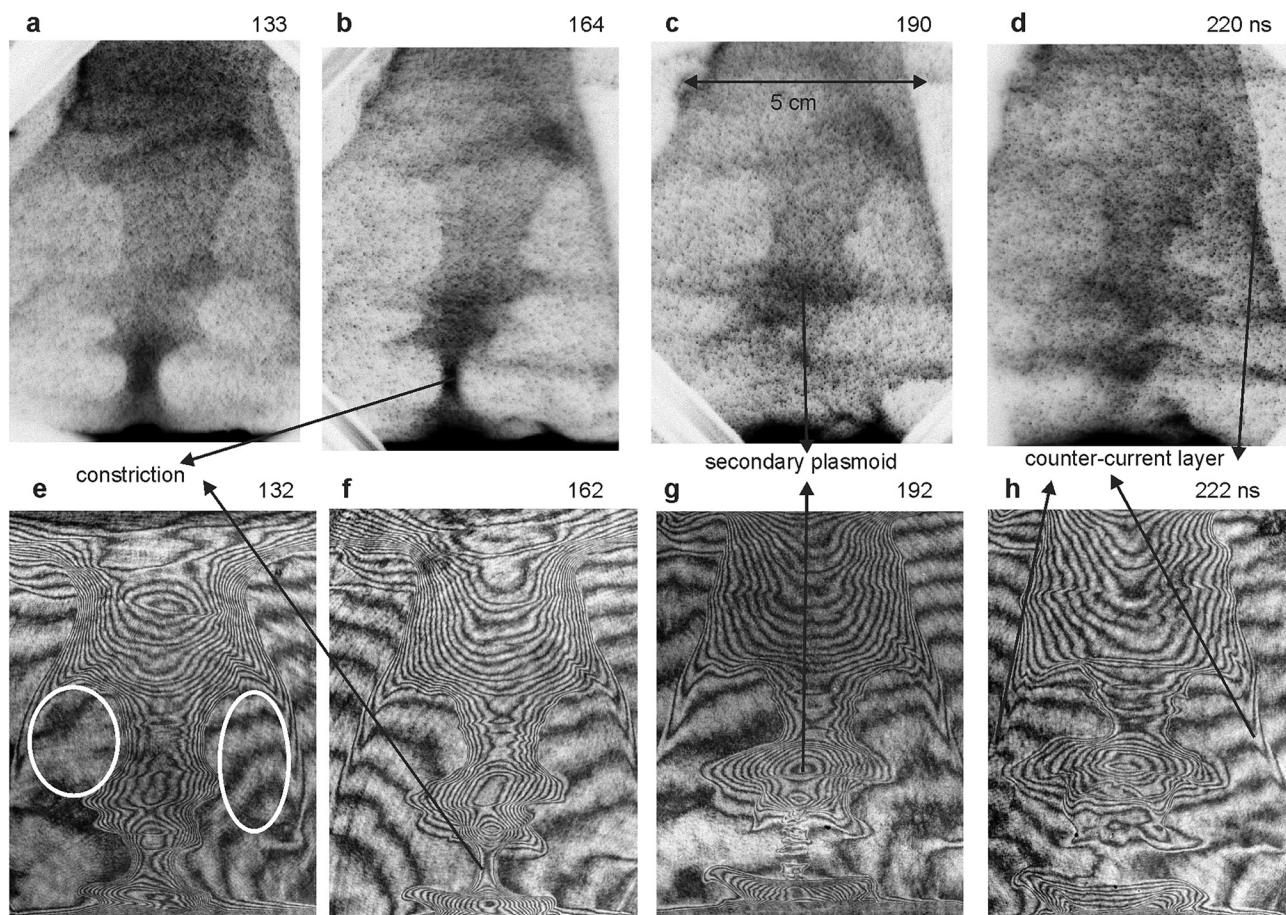


FIG. 9. Data from the shot #12597: the XUV [(a)–(d)] and interferometric [(e)–(h)] frames which show the disruption of the constriction, the formation of the secondary plasmoids, the position of the counter-current layer, and the position of the toroidal tube with the quasi-stable external boundary (white mark).

a function of the radial distance from the z -axis. In the cross section of this double layered structure [see Fig. 6(g)], one can observe four regions of the total pressure changes [see Fig. 7(c)]. In the first external layer, the total pressure increases rapidly due to the action of the pinching discharge current. In the second region, inside the toroidal-like plasma structure, the pressure increases with the decreasing radius (as $1/r^2$), due to the magnetic field of the current flowing upon this structure boundary. In the third region, the surface of the dense column, one can observe the fast decrease in the total pressure. In this region, one can observe the opposite pressure gradient, what suggests that the magnetic force density ($\mathbf{j} \times \mathbf{B}$) changes its sign in a thin layer between two distinct boundaries. The fourth region, which corresponds to the interior of the dense plasma column, is characterized by the low plasma pressure. Its value is determined by the difference of the pinching poloidal component and the repulsive pressure of the plasma and toroidal current component flowing inside the toroidal-like structure. The described situation can happen when the considered current flows in the closed loop and has the opposite axial component at the inner boundary. The magnetic force density directed outward upon the inner surface of the outer structure also indicates the appearance of the toroidal current. In the toroid-like plasma structure, the similar values of the poloidal and toroidal magnetic fields could constitute very important conditions for its stability, which can exist (as one can see in the recorded pictures) during hundreds ns.

IV. SUMMARY AND CONCLUSIONS

This paper summarizes the characteristics of the closed currents surrounding the dense plasma column. The main conclusions can be formulated as follows:

- (i) In the XUV and interferometric frames, one can distinguish two outstanding boundaries: the internal boundary upon the surface of the dense pinch column, and the external one extending from the tops of the plasma lobules toward the anode. The external layer, having a smooth surface, can be considered as a region of two currents: the external current of the value and orientation of the discharge current, and the internal closed current flowing at the surface of the toroid-like plasma structure, and having poloidal and toroidal components. The magnetic field of the closed currents could generate the pressure which transported the pinching pressure from the discharge current to the dense plasma column. It contained magnetic energy in the surrounding of this column.
- (ii) The appearance of the closed currents was mentioned in earlier papers^{21,23} during the proposed interpretation of a plasma outflow from the pinch column, and the existence of some local sources of the fast deuterons in the region outside the dense plasma column. In this paper, the characteristics of the considered currents have been determined on the basis a plasma density profile recorded during the stagnation phase. It has been done at the assumption of the quasi-stationary conditions because we could neglect the kinetic energy density in comparison with the plasma pressure. The closed current (with poloidal and toroidal components) could flow along the surface of the toroidal-like plasma structure, in which the pinching pressure of the toroidal current component (equal to about 680 kA)

balanced the repulsive pressure of the poloidal current component (equal to about 270 kA). The recorded XUV frames showed a higher radiation from the considered current layer and indicated its appearance also in the region of a lower plasma density. The partial transparency of the considered layer confirmed that it has a small thickness. The appearance of the closed current in the toroidal plasma structure made it possible to store some magnetic energy in the surrounding of the dense pinch column. That energy could later be released during the acceleration of the electron and ion beams. The existence of the considered counter currents fits to the model of magnetic reconnections, which probably provide mechanisms for the acceleration of the fast ions and electrons.

The magnetic structure deduced from the XUV and interferometer images is beyond the measuring capabilities of the existing diagnostic techniques, such as Faraday rotation and magnetic probe measurements. The reason is that the Faraday rotation measurements require big dimensions, and that the polarization foils have insufficient sensitivity. Magnetic probes cannot be applied because they cannot survive the discharge, if they are placed at positions near the considered plasma structures. The described method of the determination of the characteristics of the local currents and magnetic fields, although it is based on some assumptions and conjectures, seems to be the only technique which can explain some features of the observed closed plasma structures. The role of these structures, in the storage of the magnetic energy and the formation of local sources of the fast deuterons, needs still to be explored by further experiments and theoretical studies.

It should be noted that a theoretical model of the distribution of magnetic fields, which can be associated with the spontaneously self-organized plasma structures, has just been published in a new paper,²⁸ and the results obtained with the PF-1000 facility and other PF devices are usually discussed during the Annual Meetings of the International Center of Dense Magnetized Plasmas.²⁹

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DATA AVAILABILITY

The raw data were collected at the PF-1000 large-scale facility. The derived data, supporting the findings of this study, are available from the corresponding author upon request.

REFERENCES

- ¹A. Bernard, H. Bruzzone, P. Choi, H. Chaqui, V. Gribkov, J. Herrera, K. Hirano, A. Krejci, S. Lee, C. Luo, F. Mezzetti, M. Sadowski, H. Schmidt, K.

- Ware, C. S. Wong, and V. Zaita, "Scientific status of the dense plasma focus," *J. Moscow Phys. Soc.* **8**, 93 (1998).
- ²L. Bertalot, H. Herold, U. Jager, A. Mozer, T. Oppenlander, M. Sadowski, and H. Schmidt, "Mass and energy analysis and space-resolved measurements of ions from plasma focus devices," *Phys. Lett. A* **79**, 389 (1980).
- ³A. Mozer, M. Sadowski, H. Herold, and H. Schmidt, "Experimental studies of fast deuterons, impurity- and admixture-ions emitted from a plasma focus," *J. Appl. Phys.* **53**, 2959 (1982).
- ⁴M. G. Haines, "A review of the dense Z-pinch," *Plasma Phys. Controlled Fusion* **53**, 093001 (2011).
- ⁵D. D. Ryutov, M. S. Derzon, and M. K. Matzen, "The physics of fast pinches," *Rev. Mod. Phys.* **72**, 167 (2000).
- ⁶D. D. Ryutov, "Characterizing the plasmas of dense Z-pinches," *IEEE Trans. Plasma Sci.* **43**, 2363 (2015).
- ⁷L. Soto, "New trends and future perspectives on plasma focus research," *Plasma Phys. Controlled Fusion* **47**, A361 (2005).
- ⁸V. Krauz, "Progress in plasma focus research and applications," *Plasma Phys. Controlled Fusion* **48**, B221 (2006).
- ⁹M. Krisnan, "The dense plasma focus: A versatile dense pinch for diverse applications," *IEEE Trans. Plasma Sci.* **40**, 3189 (2012).
- ¹⁰D. Klir, A. V. Shishlov, V. A. Kokshenev, P. Kubes, K. Rezac, R. H. Cherdizov, J. Cikhardt, B. Cikhardtova, D. N. Dudkin, F. I. Fursov, T. Hyhlik, J. Kaufman, B. M. Kovalchuk, J. Krasa, J. Kravarik, N. E. Kurmaev, A. Yu. Labetsky, V. Munzar, H. Orcikova, V. N. Padalko, N. A. Ratakhin, O. Sila, J. Stodulka, K. Turek, V. A. Varlachev, and R. Wagner, "Ion acceleration mechanism in mega-ampere gas-puff mechanism," *New J. Phys.* **20**, 053064 (2018).
- ¹¹V. Krauz, K. Mitrofanov, M. Scholz, M. Paduch, L. Karpinski, E. Zielinska, and P. Kubes, "Experimental study of the structure of the plasma-current sheath on the PF-1000 facility," *Plasma Phys. Controlled Fusion* **54**, 025010 (2012).
- ¹²V. I. Krauz, K. N. Mitrofanov, M. Scholz, M. Paduch, P. Kubes, L. Karpinski, and E. Zielinska, "Experimental evidence of existence of the axial magnetic field in a plasma focus," *Europhys. Lett.* **98**, 45001 (2012).
- ¹³V. I. Krauz, K. N. Mitrofanov, V. V. Myalton, V. P. Vinogradov, Yu. V. Vinogradova, E. V. Grabovsky, G. G. Zukakishvili, V. S. Koidan, and A. N. Mokeev, "Magnetic probe measurements of the current sheath on the PF-3 facility," *Plasma Phys. Rep.* **36**, 937 (2010).
- ¹⁴P. Kubes, M. Paduch, T. Pisarczyk, M. Scholz, D. Klir, J. Kravarik, K. Rezac, T. Chodukowski, I. Ivanova-Stanik, L. Karpinski, E. Zielinska, K. Tomaszewski, and M. J. Sadowski, "Experimental study of the structure of the plasma-current sheath on the PF-1000 facility," *IEEE Trans. Plasma Sci.* **38**, 672 (2010).
- ¹⁵P. Kubes, M. Paduch, T. Pisarczyk, M. Scholz, T. Chodukowski, D. Klir, J. Kravarik, K. Rezac, I. Ivanova-Stanik, L. Karpinski, M. J. Sadowski, K. Tomaszewski, and E. Zielinska, "Spontaneous transformation in the pinched column of the plasma focus," *IEEE Trans. Plasma Sci.* **39**, 562 (2011).
- ¹⁶P. Kubes, M. Paduch, M. J. Sadowski, J. Cikhardt, B. Cikhardtova, D. Klir, J. Kravarik, K. Rezac, V. Munzar, E. Zielinska, E. Skladnik-Sadowska, A. Szymaszek, K. Tomaszewski, and D. Zaloga, "Evolution of the pinched column during hard x-ray and neutron emission in a dense plasma focus," *J. Fusion Energy* **38**, 490 (2019).
- ¹⁷P. Kubes, M. Paduch, M. J. Sadowski, J. Cikhardt, B. Cikhardtova, D. Klir, J. Kravarik, V. Munzar, K. Rezac, E. Skladnik-Sadowska, A. Szymaszek, K. Tomaszewski, D. Zaloga, and E. Zielinska, "Evolution of a pinch column during the acceleration of fast electrons and deuterons in a plasma focus discharge," *IEEE Trans. Plasma Sci.* **47**, 339 (2019).
- ¹⁸P. Bellan, *Spheromaks* (Imperial College Press, London, 2000).
- ¹⁹M. A. Leontovich, "Equilibrium of plasma in magnetic field," in *Voprosy Teorii Plazmy*, edited by V. D. Shafranov (Gosatomizdat, 1963), Tom 2 (in Russian).
- ²⁰P. Kubes, M. Paduch, J. Cikhardt, B. Cikhardtova, D. Klir, J. Kravarik, K. Rezac, E. Zielinska, M. J. Sadowski, A. Szymaszek, K. Tomaszewski, and D. Zaloga, "Transformation of the ordered internal structures during the acceleration of fast charged particles in a dense plasma focus," *Phys. Plasmas* **24**, 072706 (2017).
- ²¹P. Kubes, M. Paduch, J. Cikhardt, D. Klir, J. Kravarik, K. Rezac, J. Kortanek, B. Cikhardtova, and E. Zielinska, "Existence of a return direction for plasma escaping from a pinched column in a plasma focus discharge," *Phys. Plasmas* **22**, 052706 (2015).
- ²²P. Kubes, M. Paduch, M. J. Sadowski, J. Cikhardt, B. Cikhardtova, D. Klir, J. Kravarik, V. Munzar, K. Rezac, E. Zielinska, E. Skladnik-Sadowska, A. Szymaszek, K. Tomaszewski, and D. Zaloga, "Characterization of fast deuterons involved in the production of fusion neutrons in a dense plasma focus," *Phys. Plasmas* **25**, 012712 (2018).
- ²³P. Kubes, M. Paduch, M. J. Sadowski, J. Cikhardt, B. Cikhardtova, D. Klir, J. Kravarik, R. Kwiatkowski, V. Munzar, K. Rezac, E. Skladnik-Sadowska, A. Szymaszek, K. Tomaszewski, D. Zaloga, and E. Zielinska, "Features of fast deuterons emitted from plasma focus discharges," *Phys. Plasmas* **26**, 032702 (2019).
- ²⁴K. Rezac, D. Klir, P. Kubes, and J. Kravarik, "Improvement of time-of-flight methods for reconstruction of neutron energy spectra from D(d,n)3He fusion reactions," *Plasma Phys. Control. Fusion* **54**, 105011 (2012).
- ²⁵E. Zielinska, M. Paduch, and M. Scholz, "Sixteen-frame interferometer for a study of a pinch dynamics in PF-1000 device," *Contrib. Plasma Phys.* **51**, 279 (2011).
- ²⁶V. I. Krauz, M. Paduch, K. Tomaszewski, K. N. Mitrofanov, A. M. Kharrasov, A. Szymaszek, and E. Zielinska, "Generation of compact plasma objects in plasma focus discharge," *EPL* **129**, 15003 (2020).
- ²⁷W. H. Bostic, "The pinch effect revisited," *Int. J. Fusion Energy* **1**, 1 (1977).
- ²⁸S. Auluck, "Acceleration and trapping of fast ions in selforganized magneto-plasma structures in the dense plasma focus," *Phys. Plasmas* **27**, 022308 (2020).
- ²⁹See www.icdmp.pl for the International Center for Dense Magnetized Plasma.